

Raja Rajeshwari Ganamukkula

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Summary

Senior Java Core Stack Developer with 5+ years of hands-on experience designing, developing, and maintaining enterprise-grade applications using Core Java (Java 8/11/17), Spring Boot, and microservices architecture. Deep expertise in OOP (Object-Oriented Programming), multithreading and concurrency, JVM memory management, Hibernate/JPA, and RESTful API development. Proven track record delivering scalable, high-performance, and secure backend services integrated with Oracle, PostgreSQL, and SQL Server databases across AWS and Azure cloud environments using CI/CD pipelines, Docker, and Kubernetes in Agile/Scrum teams.

Technical Skills

Core Java: Java 8/11/17, OOP (Object-Oriented Programming), multithreading and concurrency, Collections Framework, exception handling, JVM memory management, design patterns, garbage collection tuning

Frameworks and Backend: Spring Framework, Spring Boot, Spring MVC, Spring Security, Hibernate, JPA (Java Persistence API), RESTful Web Services, microservices architecture, API Gateway

Frontend and Web Technologies: Angular, React, JavaScript, HTML5, CSS3, TypeScript

Databases: Oracle, SQL Server, PostgreSQL, MySQL, strong SQL and query optimization, Entity Framework, LINQ

DevOps and Tools: Git, GitHub, GitLab, Jenkins, GitHub Actions, Maven, Gradle, Docker, Kubernetes, CI/CD (Continuous Integration and Continuous Delivery) pipelines, Blue-Green deployments, GitOps

Cloud and Monitoring: AWS (Amazon Web Services), Azure (Microsoft Azure), Azure Monitor, Log Analytics, logging and monitoring tools, Prometheus, Grafana

Event-Driven Architecture: Kafka, RabbitMQ, event-driven systems, asynchronous messaging

Engineering Practices: TDD (Test-Driven Development), unit and integration testing, secure coding standards, code reviews, Agile/Scrum, performance tuning and scalability optimization

Experience

Senior Software Engineer

07/2023 – Present

PHX Agile LLC – Client: Provident Bank

Iselin, NJ

- Designed, developed, and maintained scalable enterprise-grade applications in Core Java 17 and Spring Boot, delivering 9 production microservices processing 5M+ daily financial transactions with sub-50ms API response and 99.9% uptime.
- Developed robust RESTful APIs using Spring MVC and Spring Security with OAuth2 authentication, enabling 6 downstream integration teams to consume financial services without direct database access across enterprise API Gateway.
- Implemented multithreading and concurrency patterns using Java's Collections Framework and ExecutorService, optimizing thread pool configurations that reduced CPU contention and sustained throughput under 3x peak concurrent load.
- Optimized JVM memory management and garbage collection tuning for high-throughput Spring Boot services, profiling heap allocation hotspots and restructuring object lifecycles to eliminate full GC pauses during business-critical processing windows.
- Integrated applications with Oracle and PostgreSQL databases using Hibernate and JPA, writing optimized JPQL and native SQL queries with proper indexing strategies that reduced average query execution time across 500M+ row production tables.
- Built Kafka event-driven architecture for asynchronous financial event processing, decoupling 5 upstream producers from downstream consumers and enabling horizontal scaling of event processing without service coupling.
- Worked closely with DevOps teams to implement Jenkins and GitHub Actions CI/CD pipelines with Docker containerization and Kubernetes deployments on AWS, enabling zero-downtime Blue-Green deployment releases for all production services.

Software Engineer

01/2023 – 12/2023

Claro Software Solutions – Client: Larsen & Toubro

Hyderabad, India

- Built enterprise backend services using Spring Boot and microservices architecture for an industrial workflow management platform, delivering clean, maintainable Java code with OOP design patterns and SOLID principles across 6 independent services.
- Designed and developed RESTful Web Services with Spring MVC integrating Oracle and MySQL databases via Hibernate/JPA, applying SQL query optimization and schema indexing that sustained 50,000 database writes per minute at peak load.
- Implemented Spring Security authentication and authorization layers with role-based access control, enforcing secure coding standards across all API endpoints and passing enterprise security review with zero critical findings.
- Participated in full software development lifecycle activities — requirement analysis, design, coding, testing, deployment,

and production support — delivering sprint commitments across 12 consecutive Agile/Scrum sprints.

- Developed unit and integration test suites using JUnit and Mockito for Spring Boot services, achieving 82% code coverage across critical business logic and reducing sprint regression defects from 14 to 2 per release cycle.
- Integrated RabbitMQ event-driven messaging for asynchronous inter-service communication, decoupling 4 core workflow services and enabling independent scaling of high-load processing components without downstream impact.

Associate Consultant

Virtusa – Client: JP Morgan Chase

04/2022 – 01/2023

Hyderabad, India

- Developed high-performance Core Java (Java 11) and Spring Boot backend services for financial data processing, handling 5M+ daily trade events with multithreading concurrency patterns, Collections Framework optimization, and exception handling for resilient execution.
- Architected microservices decomposition of a legacy monolithic application, defining RESTful API contracts, shared Spring libraries, and database integration patterns that enabled 4 independent teams to develop and deploy services in parallel.
- Performed root-cause analysis and performance tuning for production JVM issues — memory leaks, thread deadlocks, and GC overhead — resolving 6 critical production incidents and documenting fixes in technical runbooks for operations teams.
- Integrated enterprise applications with SQL Server and PostgreSQL using Hibernate ORM (Object-Relational Mapping) with JPQL query optimization, maintaining consistent sub-100ms database response under sustained enterprise-scale load.
- Collaborated with architects and technical leads to implement scalable and secure application solutions on AWS, contributing to cloud-native application design reviews and enforcing security best practices across all API Gateway-exposed services.

Software Engineer

Zerocode Innovations Pvt Ltd – Client: Apollo Hospitals

09/2019 – 12/2021

Hyderabad, India

- Built and maintained enterprise-grade healthcare applications using Core Java 8, Spring Framework, and Spring Boot, supporting clinical workflow management for 500+ concurrent users across 3 hospital campuses with 99.9% application uptime.
- Developed RESTful APIs with Spring MVC and Hibernate/JPA integrated with Oracle databases, implementing strong SQL query optimization and schema design that sustained high-frequency clinical data writes and complex reporting queries at peak load.
- Applied AI/ML integration patterns to clinical decision support workflows, connecting Spring Boot backend services to ML inference APIs and delivering intelligent data processing features that reduced manual clinical triage workload.
- Supported production releases, environment management, and deployment validation activities on Azure, maintaining technical documentation, deployment procedures, and application support guides for 8 healthcare platform services.

Projects

Enterprise Microservices Platform – Core Java and Spring Boot Java 17, Spring Boot, Spring Security, Hibernate, Kafka, Oracle, Docker, Kubernetes, AWS

- Designed and delivered a scalable enterprise microservices platform in Core Java 17 and Spring Boot processing 10M+ daily transactions, with RESTful APIs, Spring Security OAuth2, Hibernate/JPA Oracle integration, and Kubernetes-based Blue-Green deployments achieving zero-downtime releases across production.
- Implemented JVM memory management optimization, Kafka event-driven architecture, and Jenkins/GitHub Actions CI/CD pipelines — applying OOP design patterns, SOLID principles, and TDD with 85%+ test coverage — establishing scalable engineering standards across a 6-service distributed system.

Performance Tuning and Scalability Optimization – Spring Boot and JVM

Java 11/17, Spring Boot, JVM

Profiling, PostgreSQL, Prometheus, Grafana, Docker

- Built a JVM performance profiling and scalability optimization framework for Spring Boot microservices, identifying multithreading contention, heap allocation hotspots, and slow Hibernate query patterns across 5 production services and resolving issues that sustained consistent sub-50ms P99 (99th percentile) latency under 5x load increase.
- Designed PostgreSQL schema optimization and SQL query tuning strategy applying indexing, partitioning, and JPQL rewrite patterns, reducing average complex query execution time and supporting a 3x increase in concurrent user load with no degradation.

Education

Master of Science, Computer Science

University of Central Missouri, Lees Summit, MO

Bachelor of Technology, Computer Science

Jawaharlal Nehru Technological University, Hyderabad, India